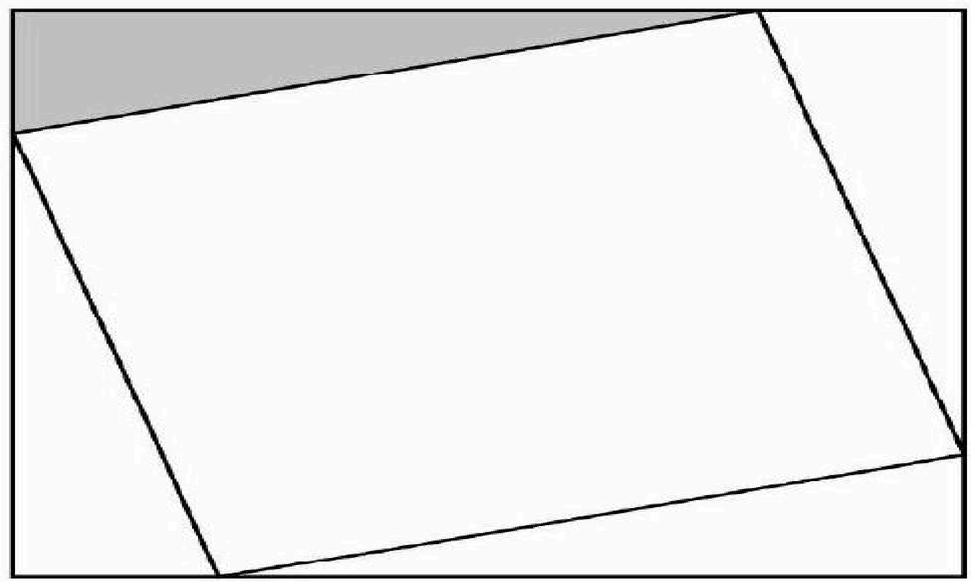
2023 AMC 10A Problem 11 - Solution

Review the full statement and step-by-step solution for 2023 AMC 10A Problem 11. Great practice for AMC 10, AMC 12, AIME, and other math contests

2023 AMC 10A Problem 11

Problem:

A square of area 2 is inscribed in a square of area 3, creating four congruent triangles, as shown below. What is the ratio of the shorter leg to the longer leg in the shaded right triangle?



Answer Choices:

A. $\frac{1}{5}$

B. $\frac{1}{4}$

C. $2-\sqrt{3}$

D. $\sqrt{3}-\sqrt{2}$

E. $\sqrt{2}-1$

Solution:

Let x be the length of the shorter leg of the right triangle. Then the longer leg has length $\sqrt{3}-x$. The hypotenuse has length $\sqrt{2}$. By the Pythagorean Theorem,

$$x^2+(\sqrt{3}-x)^2=2$$

This equation simplifies to $2x^2-2\sqrt{3}x+1=0$, and the Quadratic Formula gives the solutions $\frac{1}{2}(\sqrt{3}\pm 1)$. The requested ratio is

$$\frac{\frac{1}{2}(\sqrt{3}-1)}{\frac{1}{2}(\sqrt{3}+1)}=\text{(E)}\boxed{2-\sqrt{3}}$$